

Reconciling the DiD-BCF Theory with the Calibration Experiments

Hugo Gobato Souto

September 12, 2026

Abstract

The published JBES simulation grid and the new `Extra_Theory_Experiments` (ETE) calibration runs share the structured DiD-BCF sampler and the same canonical data-generating processes, so they can be compared directly. This note maps the published estimator labels onto the new implementations, reproduces the published numbers from the archived per-replication summaries, isolates the one pathology the published paper flags (the doubly robust correction explodes at nonlinear degrees), and answers two questions: whether the theory-aligned estimator improves on the old one, and whether it remains competitive with the published benchmarks. The explosion is a posterior *tail* produced by the unregularized logistic propensity pilot under covariate separation, not a property of the structured sampler or of the correction algebra. At degree $d = 2$ and $N = 200$ two of one hundred replications carry errors 53.6 and 25.5 in the same-sample correction, and four of one hundred carry errors up to 68.5 in the reference fold convolution with the logit pilot; removing those replications returns every RMSE to about 0.29. The same construction with a random-forest, intercept, or clipped pilot has no tail, and the `oracle_canonical` design with exact bounded propensities is stable throughout. Against the old corrected estimator, the theory-aligned version is about the same at $d = 1$ and under AR(1) errors, and strictly better at $d = 2$ and at $N = 800$; against the benchmarks, the raw structured fit remains the most accurate estimator on the common estimands, and the theory-aligned version is competitive with Callaway–Sant’Anna and DoubleML and much better calibrated than grf-DiD, though it does not dominate them on serial-correlation point accuracy. The raw fit’s better point accuracy is not evidence against the correction: the theory makes no root- n coverage claim for the raw summary, and the raw intervals undercover under serial correlation at every sample size, which is exactly the failure the correction repairs. All numbers come from the now complete calibration archives (48 of 48 correction shards, 48 of 48 information shards) and the published revision summaries, through a single reproducible script.

1 Purpose and provenance

The JBES submission evaluates DiD-BCF with a simulation grid whose plain and corrected summaries were produced by the revision code base (`Simulation_Studies_Revision`). The ETE package reruns the same canonical and staggered data-generating processes through the structured sampler but replaces the single corrected estimator of that grid by a family: the raw structured posterior, the same-sample correction reimplemented locally as `current_hybrid`, the theorem-aligned fixed- K reference fold convolution of Algorithm 2, and the two oracle variants on an oracle-ready design. The purpose of this note is to state precisely what matches, what changed, and why, and to compare the new estimators with both the published DiD-BCF numbers and the published frequentist benchmarks.

Two provenance facts matter for reading everything below. First, the simulation grid in the submission already uses the structured sampler, so the reconciliation here concerns the post-processing step, not the sampler; the seed instability documented for the `mpdta` application concerns

the older **published** specification and is a separate, already documented issue. Second, the ETE runs were budgeted at the production sampler settings (`num_gfr` = 50, `num_mcmc` = 500, `keep_every` = 5, `num_chains` = 3), so differences from the published grid are not budget artifacts.

Table 1 maps the estimator labels used in the two code bases.

Table 1: Estimator map. “Reproduced” means the ETE run uses the same construction as the published grid up to seeds and fold draws.

Published label	ETE label	Construction
DiD-BCF (plain)	<code>raw_structured</code>	Average of raw structured posterior τ draws over treated units in the cell.
DiD-BCF (corrected)	<code>current_hybrid</code> with <code>logit</code> , <code>intercept</code> , or <code>rf</code> , optional clipping	Algorithm 1 on the same-sample posterior: Bayesian-bootstrap weights, efficient-influence-function augmentation using a cross-fitted pilot propensity, and a posterior bias correction using pilot m and π .
(new)	<code>reference_fold_convolution</code> with <code>logit</code> , <code>rf</code> , or <code>intercept</code>	Algorithm 2: fixed- K folds of original units, off-fold pilots, a fold-specific posterior fitted on the evaluation fold, and a size-weighted convolution of fold draws.
(new)	<code>oracle_current_hybrid</code> , <code>oracle_reference_fold_convolution</code>	The two corrections run with exact <code>m0_oracle</code> , <code>pi_oracle</code> , and <code>barpi_oracle</code> on the locally added <code>oracle_canonical</code> design whose propensity is Bernoulli with a known bounded probability and whose population truth is exactly 3.

2 Runs, estimators, and metric conventions

All errors below use replication-specific truths: for replication r , $\text{err}_r = \hat{\theta}_r - \theta_{0,r}$, $\text{SD}(\text{err})$ is the Monte-Carlo standard deviation of those errors, RMSE their root mean square, `Cov.95` the frequency with which the reported $[q_{0.025}, q_{0.975}]$ interval contains $\theta_{0,r}$, `Len.95` its average width, and $\overline{\text{sd}}/\text{SD}$ the average reported posterior standard deviation divided by $\text{SD}(\text{err})$. The published tables divided by the raw spread of estimates instead, so the ratio column here is *recomputed* on the common error scale; bias, $\text{SD}(\text{err})$, RMSE , coverage, and length reproduce the published tables exactly. Rejection rates use the two-sided 5% rule that generated the published null table, namely $\min\{P(\tau > 0), P(\tau \leq 0)\} < 0.025$.

Both families are now complete. The correction family has all 48 shards and the information family has all 48 shards, so every cell has 100 replications; the 2,400 oracle requests on the production DGPs remain explicitly “unavailable” by design, as they should, because those DGPs do not expose exact nuisances. Monte-Carlo standard errors on a bias of magnitude 0.1 with $\text{SD}(\text{err}) \approx 0.1$ are about 0.010, rising to 0.025 when $\text{SD}(\text{err}) = 0.25$; coverage rates have Monte-Carlo standard errors of at most 0.05 at $p = 0.5$, and of about 0.022 at $p = 0.05$. A difference of 0.02 to 0.03 between an old and a new number is therefore at the edge of resolution, and the qualitative patterns below are far larger than that.

3 Reproduction of the published results

Tables 2 and 3 compare the published grid with the new runs for the cell that every estimator reports, $\text{GATT}(4, 4)$, at $N = 200$ and degrees $d = 1, 2$. Table 4 repeats the comparison for the

pooled ATT that carries the main-text table of the submission, and Table 5 extends the comparison to $N = 800$.

The reproduction is close where it matters. At $d = 1$ the new raw structured fit matches the published plain fit closely, for instance baseline bias -0.018 against -0.022 and $\text{SD}(\text{err})$ 0.131 against 0.109 (different replication panels), and the new `current_hybrid` matches the published corrected fit, for instance serial coverage 0.980 against 0.990 and length 0.696 against 0.693 . At $N = 800$ the new raw fit tracks the published plain fit within 0.011 on RMSE in all four cells of Table 5, and the new corrected variants match the direction and roughly the magnitude of the published correction.

The nonlinear corner is the interesting one. At baseline $d = 2$, $N = 200$, the published corrected fit has $\text{RMSE} = 1.040$ against a plain-fit $\text{RMSE} = 0.122$; the same-sample correction in the new run has $\text{RMSE} = 5.947$, and the reference fold convolution with the logit pilot has $\text{RMSE} = 6.894$. The raw fit is unaffected ($\text{RMSE} = 0.135$). The next section shows that these large numbers are carried by a handful of replications in both runs, which is the sense in which the published “instability at nonlinear degrees” is reproduced rather than contradicted.

4 The nonlinear-degree explosion is a propensity-pilot tail

The per-replication errors settle the question. In the published baseline $d = 2$, $N = 200$ corrected run, two of 100 replications have $|\text{err}| > 2$ (the largest is 9.41) and deleting those two reduces RMSE from 1.040 to 0.240 , which is the raw scale. In the new same-sample logit run, two of 100 replications exceed 2 (the largest is 53.63) and deleting them reduces RMSE from 5.947 to 0.294 ; in the reference fold logit run, four of 100 exceed 2 (the largest is 68.52) and deleting them reduces RMSE from 6.894 to 0.292 . The median absolute errors are 0.159 for the published fit and 0.177 to 0.190 for the two new logit fits, so the central behavior of all three is the same and only the tail differs.

The diagnostics identify the cause. In the same-sample logit run at that cell the largest observed control odds is at the clipping bound, $\hat{\pi}/(1 - \hat{\pi}) = 999$, the effective sample size falls to 1.02 , and the maximum normalized control weight reaches 0.988 ; in the reference fold logit run the maximum odds are 986 with minimum effective sample size 1.15 . The $d = 2$ covariates are Kang–Schafer transforms including $e^{X_4/2}$ and cubic terms, so an unregularized logistic fit separates the groups, the inverse-odds factor in the augmentation and in $\hat{\gamma}$ becomes enormous, and a single control observation can dominate the corrected draw. The failure is a violation of the bounded-overlap and localization conditions of the BvM theorem (Conditions on allocation and rates in Section S7 of the theory supplement require $1 - \hat{\pi}$ and $\hat{\gamma}$ to stay in fixed L^∞ balls), so the observed tail lies outside the theorem rather than contradicting it.

Three pieces of evidence confirm the diagnosis. First, capping the odds at 19 by setting the clipping constant to 0.05 eliminates the tail: in the same cell the maximum $|\text{err}|$ falls from 53.63 to 0.770 , RMSE falls from 5.947 to 0.258 , coverage is 0.980 , and the average length is 1.062 . Second, replacing the logit pilot by the random-forest pilot, whose odds range from a median of 2.5 to a maximum of 5.3 with minimum effective sample size 33.1 , removes the tail without any clipping: maximum $|\text{err}| = 0.839$ and $\text{RMSE} = 0.234$, with coverage 0.970 and length 0.986 . The intercept pilot behaves similarly ($\text{RMSE} = 0.214$). Third, Table 6 reports the `oracle_canonical` design, in which the propensity is known to lie in $(0.001, 0.999)$ and m_0 is exact. There the $d = 2$, $N = 200$ corrected variants all stay between $\text{RMSE} = 0.199$ (current RF) and 0.265 (reference logit), with no tail at all, and at $N = 800$ every corrected variant has RMSE between 0.109 and 0.130 . Holding the correction fixed and improving overlap removes the pathology; holding overlap fixed and changing the fold construction does not.

5 Performance of the reference fold convolution

The reference fold convolution is the estimator the theory actually covers, so its finite-sample profile deserves a direct reading. At $d = 1$, $N = 200$, baseline, its random-forest version has bias -0.030 , $\text{SD}(\text{err}) = 0.243$, $\text{RMSE} = 0.243$, coverage 0.910, and length 0.962, against published corrected bias -0.028 , $\text{SD}(\text{err}) = 0.231$, $\text{RMSE} = 0.231$, coverage 0.950, length 0.979. It is therefore about as efficient as the published correction at that cell, with slightly wider intervals and coverage about 1.8 Monte-Carlo standard errors below nominal; its value is that it is the construction the BvM theorem describes, and with a stable pilot it remains controlled at $d = 2$ where the old published run did not ($\text{RMSE} = 0.234$ against 1.040). At $N = 800$ the reference variants track the same-sample variants closely, for example baseline $d = 1$ reference RF $\text{RMSE} = 0.130$ against published corrected 0.107 and new current logit 0.109, with coverage 0.920 against 0.980 and 0.940 and length 0.477 against 0.450.

Two honest qualifications follow. The correction, in any of its implementations, pays an efficiency cost at $N = 200$: in the baseline cells the raw RMSE is about 0.13 against 0.21 to 0.26 for the corrected variants, with intervals roughly 1.6 to 1.9 times longer, although under serial correlation at $d = 2$ the corrected variants are close to or better than the raw fit (0.145 to 0.164 against 0.141). And the reference construction does not by itself fix the pilot: with the logit pilot it inherits the tail described in Section 4. The practical recommendation is to pair the theorem-aligned fold construction with a pilot that respects overlap, which the random-forest pilot does out of the box in these runs, or to report clipping explicitly as an exploratory stabilization.

6 Is the theory-aligned estimator better than the old corrected one?

Table 13 puts the two corrections side by side in the cells where they matter. The answer is that it is about the same in the well-behaved cells and strictly better in the cells that used to fail, which is the same as saying that the theory-aligned construction removes the failure without giving up what worked.

At $d = 1$ and $N = 200$, baseline, the published corrected RMSE is 0.231 with coverage 0.950 and the new reference RF RMSE is 0.243 with coverage 0.910; under serial correlation at $d = 1$ the numbers are 0.136 with coverage 0.990 against 0.160 with coverage 0.960. Those differences are small and within about one to two Monte-Carlo standard errors, with the reference variant slightly more conservative in length and slightly lower in coverage. The same-sample logit variant is closer to the published correction in these cells (0.214 and 0.145), as expected, since it is the same algebra.

At $d = 2$ the comparison stops being close. At $N = 200$, baseline, the published corrected RMSE is 1.040; the reference RF is 0.234 with coverage 0.970, the same-sample RF is 0.224 with coverage 0.970, and the clipped logit is 0.258 with coverage 0.980. At $N = 800$, baseline, the published corrected RMSE is 0.318 against 0.114 for the reference RF and 0.102 for the same-sample logit. Under serial correlation at $d = 2$, $N = 200$, the published corrected RMSE is 0.269 against 0.164 for the reference RF.

The verdict is therefore: same performance in the benign regimes, clearly better in the nonlinear and large- N regimes, and no longer at the mercy of a handful of replications. The one caveat is that the reference construction inherits whatever the pilot does, so the improvement comes from the joint choice of the off-fold Algorithm 2 construction and an overlap-respecting pilot, not from Algorithm 2 alone.

7 Is the theory-aligned estimator still better than the benchmarks?

Tables 7, 8, and 9 compare the new estimators with the published frequentist benchmarks on the same designs and replication-specific truths, and Table 14 condenses the GATT(4, 4) comparison. The short answer is that the raw structured fit is still the accuracy leader, while the theory-aligned corrected fit is now competitive with the best benchmark rather than dominated by it.

On point accuracy at $N = 200$, baseline, $d = 1$, the raw fit has $\text{RMSE} = 0.131$ against 0.221 for Callaway–Sant’Anna, 0.304 for DoubleML, and 0.222 for grf-DiD. The same-sample correction has $\text{RMSE} = 0.214$, matching Callaway–Sant’Anna and beating DoubleML, and the reference RF has 0.243, between Callaway–Sant’Anna and DoubleML. At $d = 2$ the raw fit is again first (0.135), and the corrected versions are 0.224 to 0.258 against 0.225 for Callaway–Sant’Anna, 0.280 for DoubleML, and 0.223 for grf-DiD, so they are level with the two best benchmarks. Coverage favors the new corrected versions over grf-DiD by a wide margin (0.97 against 0.81 at both degrees) and is comparable to Callaway–Sant’Anna and DoubleML.

The one place where the corrected fit does not match the best benchmark is serial correlation. At $d = 1$, $N = 200$, the raw fit has the best RMSE (0.126) but coverage 0.920, Callaway–Sant’Anna has 0.129 with coverage 0.990, and the reference RF has 0.160 with coverage 0.960; at $d = 2$ the pattern repeats with Callaway–Sant’Anna at 0.135 against 0.164 for the reference RF. So the corrected DiD-BCF pays a visible efficiency cost on serially correlated data relative to the purpose-built efficient benchmark, while remaining close to DoubleML and clearly better calibrated than grf-DiD. At $N = 800$ the raw fit dominates every benchmark by a wide margin ($\text{RMSE} = 0.069$ against 0.107 to 0.114), and the corrected versions are at 0.109 to 0.130, in the same neighborhood as the benchmarks.

The pooled ATT tells the same story (Table 9). At baseline $d = 1$ the raw fit has $\text{RMSE} = 0.112$ against 0.179 for TWFE and 0.169 for Synthetic DiD; the same-sample correction has 0.176, essentially TWFE, and the reference RF has 0.203, a little worse than both but with coverage 0.930. Under serial correlation at $d = 1$ the raw fit’s coverage drops to 0.810 while the same-sample correction holds 0.980 at $\text{RMSE} = 0.142$, better than TWFE (0.198) and close to Synthetic DiD (0.166), which is exactly the trade the theory predicts.

In one sentence: the raw structured DiD-BCF remains the most accurate estimator on every common cell, the theory-aligned corrected version is now competitive with Callaway–Sant’Anna and DoubleML and far better calibrated than grf-DiD, and it no longer collapses at nonlinear degrees, but it does not uniformly dominate Callaway–Sant’Anna under serial correlation, where the efficient frequentist benchmark is still the sharper instrument for a point estimate.

8 Does the raw fit’s accuracy indicate that the correction is wrong?

No, and the experiments show why. “Raw” means the plain structured posterior summary, the average of the τ draws over treated units in the cell, before any post-processing. It is the estimator the paper recommends for heterogeneity analyses and point summaries, and the theory deliberately makes no root- n coverage claim for it: Section S7 states that uncorrected intervals require an additional equicontinuity regime and that raw plug-in summaries carry no root- n guarantee, and the minimax lower bound in the main text makes the complementary point that no estimator can have a uniform root- n rate for the CATT surface, so any average of it inherits either a smoothness-limited rate or an extra localization requirement.

Three features of the numbers confirm that the correction, not the raw fit, is doing its job. First, the raw intervals are the ones that fail when they should: under AR(1) errors their coverage is 0.88

to 0.92 at $N = 200$ and remains 0.89 to 0.92 at $N = 800$, a missing variance component that does not vanish with sample size, while the corrected variants cover at 0.96 to 1.00 at both sizes. A point estimator can have a low RMSE and still be uncalibrated; here it is. Second, at $N = 800$ the corrected interval lengths match the asymptotically efficient benchmark almost exactly (baseline $d = 1$: 0.451 against 0.445 for Callaway–Sant’Anna; serial $d = 1$: 0.329 against 0.326), which is the quantitative content of the BvM theorem; the raw intervals are much shorter (0.286 and 0.234) precisely because they are not calibrated to the sampling variability of the efficient average. Third, the oracle design, where the nuisance conditions of the theorem hold exactly, produces corrected RMSEs of 0.199 to 0.265 at $d = 2$, $N = 200$ with coverage 0.95 to 0.99, and no tail; if the correction algebra were wrong, exact nuisances would not rescue it.

The raw fit’s better RMSE at $N = 200$ is a bias-variance outcome. The effect-forest prior shrinks toward smooth, small effects, and these DGPs have smooth low-dimensional effect surfaces, so at small N the shrinkage wins while the correction pays the variance of estimating the nuisance pilots and the augmentation. That is a finite-sample property of a well-matched prior and a smooth truth, not a failure of an asymptotic statement, and the efficiency claim is a limiting local statement about the corrected law. The two live side by side: report the raw fit for point accuracy and heterogeneity, and the corrected fit for calibrated inference, which is exactly the paper’s recommendation.

9 Information ablation

Tables 10 and 11 report the information ablation. The reduced-cell fit uses only cohort g and never-treated units at periods $g - 1$ and t ; the pooled fit keeps the full panel but forces a single effect surface with `effect_by_cohort=False`.

For the canonical designs the reduction is viable but costly. At baseline $d = 1$, $N = 800$, the reduced cell raises GATT(4, 4) RMSE from 0.065 to 0.111 and roughly doubles the interval length from 0.288 to 0.493, with coverage rising from 0.95 to 0.98; at $N = 200$ the RMSE rises from 0.127 to 0.207 with length from 0.559 to 0.936. Under serial correlation the reduction is nearly free in RMSE at $N = 800$ (0.060 against 0.058) and slightly favorable at $N = 200$ $d = 2$ (0.126 against 0.134), always with coverage at or above nominal. The consistent pattern is that discarding the other periods and cohorts costs some efficiency and a good deal of length, not validity.

Pooling is a different matter. With a single treated cohort, baseline and serial, the pooled fit is numerically almost identical to the full-panel fit, for example baseline $d = 1$, $N = 800$, RMSE 0.067 against 0.065 and coverage 0.97 against 0.95, because there is nothing to pool across. With staggered cohorts it is catastrophic: the pooled fit reports a single surface for cohorts whose effects differ, and the GATT(4, 4) bias is 1.417 at $N = 200$ and 1.467 at $N = 800$ with coverage 0.00 in every cell, while the reduced-cell fit stays approximately unbiased (0.012 and 0.022) with coverage 1.00. The pooled ATT is similarly biased (−0.204 at $N = 800$ with coverage 0.13). The ablation therefore says that the two-group two-period reduction is a defensible local device and that pooling cohorts is not, which is the expected reading of the no-already-treated-as-controls principle.

10 Sharp-null size

Table 12 reports the two-sided 5% rejection rate under the sharp null. With the complete data the picture is cleaner than in the partial run. The raw structured fit is conservative everywhere, between 0.010 and 0.080. The published corrected numbers in the matched cells are 0.040 to 0.090, the new same-sample variants are between 0.000 and 0.070, and the reference variants are between 0.040 and 0.100. Every cell is within about 1.7 Monte-Carlo standard errors of nominal, including

the largest cells, the reference-logit ATT entries at 0.100, whose Monte-Carlo standard error is 0.030. The correction therefore neither restores exact size nor destroys it: it trades some of the raw fit’s conservativeness for better interval calibration, at nominal size.

11 What the reconciliation establishes

With all correction shards in hand the account is complete. The published plain and corrected central behavior is reproduced with the current structured sampler; the published instability of the corrected estimator at $d \geq 2$ is reproduced and localized to a small-replication tail caused by inverse odds under logistic separation; the tail exists equally in the theorem-aligned reference construction when that construction is handed the same pilot, and disappears with the random-forest, intercept, or clipped pilots, and by construction on the oracle design. The theory-aligned construction fixes the independence and weighting structure of the correction, while the overlap conditions remain an empirical obligation on the pilot. The paper’s advice to report both fits side by side remains correct, with the amendment that the corrected fit should name its pilot and that a logit pilot on transformed covariates with 10^{-3} clipping should not be the default.

The experiments here do not establish any theorem. They do not cover the contracted posterior rate, they do not have degree $d = 3$ in the correction grid, they use finite posterior and bootstrap draw budgets, and the oracle design is deliberately idealized (bounded propensity and iid row errors). They are calibration evidence about which implementation of the theory behaves as advertised.

12 Limitations

All calibration cells now have 100 replications, and the published and new runs use different replication panels, so the exact magnitudes of tail-driven quantities are replication-specific; this is why Section 4 emphasizes the diagnostic and the trimmed RMSE rather than the raw RMSE. The benchmark comparison in Section 7 uses the published benchmark runs on the same DGP designs, so it is a comparison across independent panels rather than a paired comparison. The staggered comparison between the published `D_staggered` run and the new information family is consistent in direction and roughly two Monte-Carlo standard errors apart in bias, which we read as agreement within seed allocation. Finally, clipping is an exploratory stabilization and is not theoretically neutral, so the clipped results should be read as a sensitivity check alongside the random-forest pilot, not as a theorem-covered default.

Table 2: GATT($g=4, t=4$), $N = 200$, baseline design: published JBES run (100 replications) against the Extra_Theory calibration run (available replications shown). Metrics use replication-specific truths; here $\overline{\text{sd}}/\text{SD}$ is average reported posterior SD over the Monte-Carlo error SD $\text{SD}(\hat{\theta} - \theta_0)$, which is the common scale used throughout this note (the published tables divided by the raw spread of estimates instead).

Estimator	Bias	SD(err)	RMSE	Cov.95	Len.95	$\overline{\text{sd}}/\text{SD}$	n
<i>degree $d = 1$</i>							
JBES plain (structured)	-0.022	0.109	0.110	1.000	0.558	1.305	100
JBES corrected (same-sample)	-0.028	0.231	0.231	0.950	0.979	1.082	100
New raw structured	-0.018	0.131	0.131	0.980	0.561	1.097	100
New current hybrid, logit	-0.036	0.212	0.214	0.960	0.965	1.160	100
New current hybrid, logit clip .05	-0.036	0.212	0.214	0.960	0.965	1.160	100
New current hybrid, RF	-0.029	0.208	0.209	0.980	0.937	1.146	100
New reference fold, logit	-0.031	0.254	0.254	0.910	0.994	0.998	100
New reference fold, RF	-0.030	0.243	0.243	0.910	0.962	1.010	100
New reference fold, intercept	-0.029	0.221	0.222	0.940	0.897	1.036	100
<i>degree $d = 2$</i>							
JBES plain (structured)	-0.024	0.120	0.122	0.980	0.572	1.214	100
JBES corrected (same-sample)	0.109	1.039	1.040	0.960	1.676	0.409	100
New raw structured	0.018	0.134	0.135	0.960	0.575	1.090	100
New current hybrid, logit	0.796	5.923	5.947	0.980	4.024	0.179	100
New current hybrid, logit clip .05	0.021	0.259	0.258	0.980	1.062	1.046	100
New current hybrid, RF	0.020	0.224	0.224	0.970	0.965	1.097	100
New reference fold, logit	0.651	6.898	6.894	0.920	3.546	0.141	100
New reference fold, RF	0.022	0.234	0.234	0.970	0.986	1.073	100
New reference fold, intercept	0.022	0.214	0.214	0.960	0.910	1.087	100

Table 3: GATT($g=4, t=4$), $N = 200$, AR(1) serial design: published JBES run (100 replications) against the Extra_Theory calibration run (available replications shown). Metrics use replication-specific truths; here $\overline{\text{sd}}/\text{SD}$ is average reported posterior SD over the Monte-Carlo error SD $\text{SD}(\hat{\theta} - \theta_0)$, which is the common scale used throughout this note (the published tables divided by the raw spread of estimates instead).

Estimator	Bias	SD(err)	RMSE	Cov.95	Len.95	$\overline{\text{sd}}/\text{SD}$	n
<i>degree $d = 1$</i>							
JBES plain (structured)	-0.012	0.111	0.111	0.970	0.462	1.058	100
JBES corrected (same-sample)	-0.024	0.134	0.136	0.990	0.693	1.317	100
New raw structured	0.003	0.127	0.126	0.920	0.465	0.935	100
New current hybrid, logit	-0.008	0.146	0.145	0.980	0.696	1.219	100
New current hybrid, logit clip .05	-0.008	0.146	0.145	0.980	0.696	1.219	100
New current hybrid, RF	-0.009	0.144	0.144	0.980	0.672	1.192	100
New reference fold, logit	-0.003	0.185	0.184	0.940	0.717	0.987	100
New reference fold, RF	-0.012	0.161	0.160	0.960	0.687	1.090	100
New reference fold, intercept	-0.010	0.154	0.154	0.950	0.646	1.066	100
<i>degree $d = 2$</i>							
JBES plain (structured)	-0.013	0.127	0.128	0.960	0.478	0.957	100
JBES corrected (same-sample)	0.019	0.269	0.269	0.990	1.160	1.101	100
New raw structured	-0.016	0.141	0.141	0.880	0.480	0.871	100
New current hybrid, logit	-0.025	0.264	0.264	0.990	0.969	0.935	100
New current hybrid, logit clip .05	-0.001	0.159	0.158	0.990	0.761	1.223	100
New current hybrid, RF	0.002	0.145	0.145	1.000	0.688	1.207	100
New reference fold, logit	-0.014	0.327	0.325	0.950	0.942	0.738	100
New reference fold, RF	0.003	0.165	0.164	0.980	0.701	1.083	100
New reference fold, intercept	0.003	0.147	0.147	0.990	0.655	1.137	100

Table 4: Pooled ATT, $N = 200$: published JBES run against the new calibration runs. The ratio column is recomputed on the common error-SD scale.

Estimator	Bias	SD(err)	RMSE	Cov.95	Len.95	$\overline{\text{sd}}/\text{SD}$	n
<i>baseline, degree $d = 1$</i>							
JBES plain (structured)	0.000	0.098	0.098	0.960	0.425	1.105	100
JBES corrected (same-sample)	-0.000	0.180	0.179	0.980	0.813	1.152	100
New raw structured	-0.007	0.113	0.112	0.950	0.425	0.967	100
New current hybrid, logit	-0.015	0.177	0.176	0.950	0.794	1.148	100
New current hybrid, logit clip .05	-0.015	0.177	0.176	0.950	0.794	1.148	100
New current hybrid, RF	-0.014	0.172	0.171	0.960	0.773	1.148	100
New reference fold, logit	-0.028	0.213	0.213	0.930	0.821	0.982	100
New reference fold, RF	-0.015	0.203	0.203	0.930	0.793	0.994	100
New reference fold, intercept	-0.006	0.195	0.194	0.960	0.746	0.977	100
<i>baseline, degree $d = 2$</i>							
JBES plain (structured)	-0.003	0.116	0.115	0.940	0.437	0.968	100
JBES corrected (same-sample)	0.116	1.069	1.070	0.970	1.502	0.357	100
New raw structured	0.014	0.120	0.120	0.950	0.442	0.943	100
New current hybrid, logit	0.468	4.008	4.015	0.990	2.838	0.186	100
New current hybrid, logit clip .05	0.004	0.215	0.214	0.990	0.900	1.068	100
New current hybrid, RF	0.008	0.172	0.171	0.980	0.794	1.179	100
New reference fold, logit	0.579	5.759	5.759	0.870	2.812	0.132	100
New reference fold, RF	0.022	0.180	0.180	0.970	0.820	1.161	100
New reference fold, intercept	0.024	0.161	0.162	0.970	0.761	1.209	100
<i>serial, degree $d = 1$</i>							
JBES plain (structured)	0.005	0.127	0.126	0.830	0.360	0.732	100
JBES corrected (same-sample)	-0.002	0.150	0.150	0.980	0.704	1.193	100
New raw structured	0.004	0.130	0.130	0.810	0.361	0.714	100
New current hybrid, logit	-0.010	0.143	0.142	0.980	0.692	1.237	100
New current hybrid, logit clip .05	-0.010	0.143	0.142	0.980	0.692	1.237	100
New current hybrid, RF	-0.012	0.143	0.143	0.970	0.674	1.201	100
New reference fold, logit	-0.003	0.190	0.189	0.940	0.710	0.952	100
New reference fold, RF	-0.018	0.172	0.172	0.930	0.680	1.010	100
New reference fold, intercept	-0.013	0.166	0.166	0.940	0.641	0.985	100
<i>serial, degree $d = 2$</i>							
JBES plain (structured)	0.000	0.144	0.143	0.820	0.378	0.672	100
JBES corrected (same-sample)	0.071	0.477	0.480	0.960	1.234	0.640	100
New raw structured	-0.018	0.149	0.149	0.770	0.381	0.657	100
New current hybrid, logit	-0.003	0.332	0.330	0.980	1.091	0.836	100
New current hybrid, logit clip .05	-0.011	0.179	0.179	0.980	0.746	1.064	100
New current hybrid, RF	-0.009	0.161	0.160	0.980	0.698	1.108	100
New reference fold, logit	-0.052	0.558	0.558	0.880	0.954	0.434	100
New reference fold, RF	-0.017	0.197	0.196	0.940	0.711	0.922	100
New reference fold, intercept	-0.010	0.171	0.171	0.950	0.660	0.988	100

Table 5: GATT($g=4, t=4$), $N = 800$: the published $B2$ sweep against the new calibration runs.

Estimator	Bias	SD(err)	RMSE	Cov.95	Len.95	$\overline{\text{sd}}/\text{SD}$	n
<i>baseline, degree $d = 1$</i>							
JBES plain (structured)	0.002	0.069	0.068	0.970	0.288	1.073	100
JBES corrected (same-sample)	0.001	0.107	0.107	0.980	0.450	1.072	100
New raw structured	0.008	0.069	0.069	0.960	0.286	1.059	100
New current hybrid, logit	0.004	0.109	0.109	0.940	0.451	1.058	100
New current hybrid, logit clip .05	0.004	0.109	0.109	0.940	0.451	1.058	100

Estimator	Bias	SD(err)	RMSE	Cov.95	Len.95	$\overline{\text{sd}}/\text{SD}$	n
New current hybrid, RF	0.001	0.110	0.109	0.950	0.471	1.098	100
New reference fold, logit	-0.001	0.122	0.122	0.920	0.456	0.954	100
New reference fold, RF	-0.004	0.130	0.130	0.920	0.477	0.936	100
New reference fold, intercept	0.000	0.114	0.113	0.930	0.446	1.002	100
<i>baseline, degree $d = 2$</i>							
JBES plain (structured)	-0.001	0.073	0.072	0.960	0.302	1.061	100
JBES corrected (same-sample)	-0.046	0.316	0.318	0.990	0.624	0.511	100
New raw structured	0.009	0.060	0.061	0.990	0.301	1.273	100
New current hybrid, logit	0.005	0.103	0.102	0.960	0.453	1.129	100
New current hybrid, logit clip .05	0.005	0.103	0.102	0.960	0.453	1.129	100
New current hybrid, RF	0.003	0.100	0.099	0.960	0.476	1.223	100
New reference fold, logit	0.008	0.109	0.108	0.950	0.457	1.078	100
New reference fold, RF	0.007	0.114	0.114	0.950	0.479	1.073	100
New reference fold, intercept	0.006	0.103	0.102	0.960	0.446	1.111	100
<i>serial, degree $d = 1$</i>							
JBES plain (structured)	0.003	0.067	0.067	0.900	0.235	0.896	100
JBES corrected (same-sample)	0.006	0.071	0.071	0.990	0.328	1.177	100
New raw structured	0.001	0.063	0.063	0.920	0.234	0.959	100
New current hybrid, logit	0.003	0.065	0.064	1.000	0.329	1.300	100
New current hybrid, logit clip .05	0.003	0.065	0.064	1.000	0.329	1.300	100
New current hybrid, RF	0.002	0.067	0.067	1.000	0.341	1.289	100
New reference fold, logit	0.001	0.068	0.068	1.000	0.330	1.235	100
New reference fold, RF	0.002	0.072	0.072	0.980	0.343	1.211	100
New reference fold, intercept	0.002	0.066	0.065	1.000	0.324	1.262	100
<i>serial, degree $d = 2$</i>							
JBES plain (structured)	0.000	0.071	0.071	0.910	0.249	0.898	100
JBES corrected (same-sample)	-0.014	0.176	0.176	1.000	0.431	0.634	100
New raw structured	0.001	0.074	0.074	0.910	0.252	0.872	100
New current hybrid, logit	-0.005	0.065	0.065	0.980	0.337	1.332	100
New current hybrid, logit clip .05	-0.005	0.065	0.065	0.980	0.335	1.322	100
New current hybrid, RF	-0.006	0.064	0.064	0.990	0.346	1.391	100
New reference fold, logit	-0.007	0.077	0.077	0.980	0.346	1.151	100
New reference fold, RF	-0.004	0.074	0.074	0.990	0.347	1.201	100
New reference fold, intercept	-0.006	0.062	0.062	0.990	0.327	1.337	100

Table 6: `oracle_canonical` design (known bounded $\pi(X) \in (0.001, 0.999)$, exact m_0 ; population truth = 3): the new estimators only.

Estimator	Bias	SD(err)	RMSE	Cov.95	Len.95	$\overline{\text{sd}}/\text{SD}$	n
<i>$N = 200$, degree $d = 1$</i>							
New raw structured	0.008	0.112	0.112	1.000	0.526	1.199	100
New current hybrid, logit	0.004	0.260	0.259	0.910	0.988	0.961	100
New current hybrid, RF	0.004	0.236	0.235	0.910	0.858	0.924	100
New reference fold, logit	0.033	0.323	0.323	0.890	1.028	0.807	100
New reference fold, RF	0.016	0.279	0.278	0.900	0.890	0.811	100
New oracle current hybrid	0.008	0.248	0.247	0.930	0.934	0.954	100
New oracle reference fold	0.009	0.254	0.253	0.930	0.947	0.946	100
<i>$N = 200$, degree $d = 2$</i>							
New raw structured	0.009	0.121	0.121	0.970	0.519	1.088	100
New current hybrid, logit	-0.025	0.222	0.222	0.960	1.020	1.163	100
New current hybrid, RF	-0.011	0.200	0.199	0.950	0.862	1.099	100
New reference fold, logit	-0.044	0.262	0.265	0.950	1.054	1.017	100

Estimator	Bias	SD(err)	RMSE	Cov.95	Len.95	$\overline{\text{sd}}/\text{SD}$	n
New reference fold, RF	-0.025	0.239	0.239	0.910	0.896	0.952	100
New oracle current hybrid	-0.007	0.221	0.220	0.970	0.985	1.130	100
New oracle reference fold	-0.008	0.224	0.223	0.970	1.001	1.128	100
$N = 800$, degree $d = 1$							
New raw structured	0.005	0.066	0.066	0.970	0.278	1.067	100
New current hybrid, logit	-0.007	0.144	0.143	0.920	0.478	0.849	100
New current hybrid, RF	-0.002	0.146	0.145	0.920	0.484	0.847	100
New reference fold, logit	-0.001	0.157	0.157	0.880	0.485	0.786	100
New reference fold, RF	-0.006	0.161	0.160	0.890	0.495	0.786	100
New oracle current hybrid	-0.000	0.137	0.136	0.940	0.470	0.878	100
New oracle reference fold	-0.000	0.137	0.136	0.940	0.472	0.879	100
$N = 800$, degree $d = 2$							
New raw structured	0.016	0.064	0.066	0.970	0.282	1.124	100
New current hybrid, logit	0.014	0.117	0.117	0.970	0.479	1.047	100
New current hybrid, RF	0.016	0.120	0.120	0.970	0.494	1.050	100
New reference fold, logit	0.015	0.122	0.123	0.960	0.481	1.005	100
New reference fold, RF	0.014	0.129	0.130	0.930	0.493	0.972	100
New oracle current hybrid	0.011	0.109	0.109	0.970	0.476	1.118	100
New oracle reference fold	0.011	0.109	0.109	0.970	0.477	1.119	100

Table 7: GATT($g=4, t=4$): the new DiD-BCF calibration estimators against the published JBES benchmarks (100 replications per cell, replication-specific truths).

Estimator	Bias	SD(err)	RMSE	Cov.95	Len.95
<i>baseline, $N = 200$, degree $d = 1$</i>					
JBES plain (structured)	-0.022	0.109	0.110	1.000	0.558
JBES corrected (same-sample)	-0.028	0.231	0.231	0.950	0.979
DiD-BCF raw (new)	-0.018	0.131	0.131	0.980	0.561
DiD-BCF same-sample, logit (new)	-0.036	0.212	0.214	0.960	0.965
DiD-BCF same-sample, logit clip .05 (new)	-0.036	0.212	0.214	0.960	0.965
DiD-BCF reference fold, RF (new)	-0.030	0.243	0.243	0.910	0.962
DiD-BCF reference fold, logit (new)	-0.031	0.254	0.254	0.910	0.994
Callaway-Sant’Anna	-0.031	0.220	0.221	0.970	0.902
DoubleML DR-DiD	-0.039	0.303	0.304	0.920	1.157
grf-DiD (Wang)	-0.036	0.220	0.222	0.810	0.630
<i>baseline, $N = 200$, degree $d = 2$</i>					
JBES plain (structured)	-0.024	0.120	0.122	0.980	0.572
JBES corrected (same-sample)	0.109	1.039	1.040	0.960	1.676
DiD-BCF raw (new)	0.018	0.134	0.135	0.960	0.575
DiD-BCF same-sample, logit (new)	0.796	5.923	5.947	0.980	4.024
DiD-BCF same-sample, logit clip .05 (new)	0.021	0.259	0.258	0.980	1.062
DiD-BCF reference fold, RF (new)	0.022	0.234	0.234	0.970	0.986
DiD-BCF reference fold, logit (new)	0.651	6.898	6.894	0.920	3.546
Callaway-Sant’Anna	-0.034	0.224	0.225	0.960	0.914
DoubleML DR-DiD	-0.034	0.279	0.280	0.940	1.127
grf-DiD (Wang)	-0.037	0.221	0.223	0.810	0.630
<i>serial, $N = 200$, degree $d = 1$</i>					
JBES plain (structured)	-0.012	0.111	0.111	0.970	0.462
JBES corrected (same-sample)	-0.024	0.134	0.136	0.990	0.693
DiD-BCF raw (new)	0.003	0.127	0.126	0.920	0.465
DiD-BCF same-sample, logit (new)	-0.008	0.146	0.145	0.980	0.696
DiD-BCF same-sample, logit clip .05 (new)	-0.008	0.146	0.145	0.980	0.696

Estimator	Bias	SD(err)	RMSE	Cov.95	Len.95
DiD-BCF reference fold, RF (new)	-0.012	0.161	0.160	0.960	0.687
DiD-BCF reference fold, logit (new)	-0.003	0.185	0.184	0.940	0.717
Callaway-Sant'Anna	-0.025	0.127	0.129	0.990	0.653
DoubleML DR-DiD	-0.029	0.163	0.165	0.990	0.776
grf-DiD (Wang)	-0.032	0.130	0.133	0.910	0.458
<i>serial, N = 200, degree d = 2</i>					
JBES plain (structured)	-0.013	0.127	0.128	0.960	0.478
JBES corrected (same-sample)	0.019	0.269	0.269	0.990	1.160
DiD-BCF raw (new)	-0.016	0.141	0.141	0.880	0.480
DiD-BCF same-sample, logit (new)	-0.025	0.264	0.264	0.990	0.969
DiD-BCF same-sample, logit clip .05 (new)	-0.001	0.159	0.158	0.990	0.761
DiD-BCF reference fold, RF (new)	0.003	0.165	0.164	0.980	0.701
DiD-BCF reference fold, logit (new)	-0.014	0.327	0.325	0.950	0.942
Callaway-Sant'Anna	-0.028	0.133	0.135	0.990	0.662
DoubleML DR-DiD	-0.022	0.170	0.170	0.970	0.773
grf-DiD (Wang)	-0.032	0.130	0.133	0.900	0.458

Table 8: GATT($g=4, t=4$): the new DiD-BCF calibration estimators against the published JBES benchmarks (100 replications per cell, replication-specific truths).

Estimator	Bias	SD(err)	RMSE	Cov.95	Len.95
<i>baseline, N = 800, degree d = 1</i>					
JBES plain (structured)	0.002	0.069	0.068	0.970	0.288
JBES corrected (same-sample)	0.001	0.107	0.107	0.980	0.450
DiD-BCF raw (new)	0.008	0.069	0.069	0.960	0.286
DiD-BCF same-sample, logit (new)	0.004	0.109	0.109	0.940	0.451
DiD-BCF same-sample, logit clip .05 (new)	0.004	0.109	0.109	0.940	0.451
DiD-BCF reference fold, RF (new)	-0.004	0.130	0.130	0.920	0.477
DiD-BCF reference fold, logit (new)	-0.001	0.122	0.122	0.920	0.456
Callaway-Sant'Anna	0.002	0.108	0.107	0.980	0.445
DoubleML DR-DiD	0.005	0.115	0.114	0.970	0.503
grf-DiD (Wang)	0.001	0.112	0.111	0.810	0.319
<i>baseline, N = 800, degree d = 2</i>					
JBES plain (structured)	-0.001	0.073	0.072	0.960	0.302
JBES corrected (same-sample)	-0.046	0.316	0.318	0.990	0.624
DiD-BCF raw (new)	0.009	0.060	0.061	0.990	0.301
DiD-BCF same-sample, logit (new)	0.005	0.103	0.102	0.960	0.453
DiD-BCF same-sample, logit clip .05 (new)	0.005	0.103	0.102	0.960	0.453
DiD-BCF reference fold, RF (new)	0.007	0.114	0.114	0.950	0.479
DiD-BCF reference fold, logit (new)	0.008	0.109	0.108	0.950	0.457
Callaway-Sant'Anna	0.001	0.108	0.108	0.980	0.449
DoubleML DR-DiD	-0.000	0.113	0.113	0.990	0.515
grf-DiD (Wang)	0.001	0.112	0.111	0.810	0.319
<i>serial, N = 800, degree d = 1</i>					
JBES plain (structured)	0.003	0.067	0.067	0.900	0.235
JBES corrected (same-sample)	0.006	0.071	0.071	0.990	0.328
DiD-BCF raw (new)	0.001	0.063	0.063	0.920	0.234
DiD-BCF same-sample, logit (new)	0.003	0.065	0.064	1.000	0.329
DiD-BCF same-sample, logit clip .05 (new)	0.003	0.065	0.064	1.000	0.329
DiD-BCF reference fold, RF (new)	0.002	0.072	0.072	0.980	0.343
DiD-BCF reference fold, logit (new)	0.001	0.068	0.068	1.000	0.330
Callaway-Sant'Anna	0.005	0.070	0.070	0.990	0.326
DoubleML DR-DiD	0.006	0.082	0.082	0.950	0.365

Estimator	Bias	SD(err)	RMSE	Cov.95	Len.95
grf-DiD (Wang)	0.007	0.074	0.074	0.890	0.244
<i>serial, N = 800, degree d = 2</i>					
JBES plain (structured)	0.000	0.071	0.071	0.910	0.249
JBES corrected (same-sample)	-0.014	0.176	0.176	1.000	0.431
DiD-BCF raw (new)	0.001	0.074	0.074	0.910	0.252
DiD-BCF same-sample, logit (new)	-0.005	0.065	0.065	0.980	0.337
DiD-BCF same-sample, logit clip .05 (new)	-0.005	0.065	0.065	0.980	0.335
DiD-BCF reference fold, RF (new)	-0.004	0.074	0.074	0.990	0.347
DiD-BCF reference fold, logit (new)	-0.007	0.077	0.077	0.980	0.346
Callaway-Sant’Anna	0.004	0.071	0.071	0.980	0.329
DoubleML DR-DiD	-0.004	0.076	0.076	1.000	0.366
grf-DiD (Wang)	0.007	0.074	0.074	0.880	0.244

Table 9: Pooled ATT: the new DiD-BCF calibration estimators against the published JBES benchmarks (100 replications per cell, replication-specific truths).

Estimator	Bias	SD(err)	RMSE	Cov.95	Len.95
<i>baseline, N = 200, degree d = 1</i>					
JBES plain (structured)	0.000	0.098	0.098	0.960	0.425
JBES corrected (same-sample)	-0.000	0.180	0.179	0.980	0.813
DiD-BCF raw (new)	-0.007	0.113	0.112	0.950	0.425
DiD-BCF same-sample, logit (new)	-0.015	0.177	0.176	0.950	0.794
DiD-BCF same-sample, logit clip .05 (new)	-0.015	0.177	0.176	0.950	0.794
DiD-BCF reference fold, RF (new)	-0.015	0.203	0.203	0.930	0.793
DiD-BCF reference fold, logit (new)	-0.028	0.213	0.213	0.930	0.821
TWFE	-0.024	0.178	0.179	0.980	0.837
Synthetic DiD	-0.023	0.168	0.169	1.000	0.797
<i>baseline, N = 200, degree d = 2</i>					
JBES plain (structured)	-0.003	0.116	0.115	0.940	0.437
JBES corrected (same-sample)	0.116	1.069	1.070	0.970	1.502
DiD-BCF raw (new)	0.014	0.120	0.120	0.950	0.442
DiD-BCF same-sample, logit (new)	0.468	4.008	4.015	0.990	2.838
DiD-BCF same-sample, logit clip .05 (new)	0.004	0.215	0.214	0.990	0.900
DiD-BCF reference fold, RF (new)	0.022	0.180	0.180	0.970	0.820
DiD-BCF reference fold, logit (new)	0.579	5.759	5.759	0.870	2.812
TWFE	-0.024	0.178	0.179	0.980	0.837
Synthetic DiD	-0.023	0.168	0.169	1.000	0.797
<i>serial, N = 200, degree d = 1</i>					
JBES plain (structured)	0.005	0.127	0.126	0.830	0.360
JBES corrected (same-sample)	-0.002	0.150	0.150	0.980	0.704
DiD-BCF raw (new)	0.004	0.130	0.130	0.810	0.361
DiD-BCF same-sample, logit (new)	-0.010	0.143	0.142	0.980	0.692
DiD-BCF same-sample, logit clip .05 (new)	-0.010	0.143	0.142	0.980	0.692
DiD-BCF reference fold, RF (new)	-0.018	0.172	0.172	0.930	0.680
DiD-BCF reference fold, logit (new)	-0.003	0.190	0.189	0.940	0.710
TWFE	-0.022	0.198	0.198	0.980	0.894
Synthetic DiD	-0.024	0.165	0.166	0.990	0.792
<i>serial, N = 200, degree d = 2</i>					
JBES plain (structured)	0.000	0.144	0.143	0.820	0.378
JBES corrected (same-sample)	0.071	0.477	0.480	0.960	1.234
DiD-BCF raw (new)	-0.018	0.149	0.149	0.770	0.381
DiD-BCF same-sample, logit (new)	-0.003	0.332	0.330	0.980	1.091

Estimator	Bias	SD(err)	RMSE	Cov.95	Len.95
DiD-BCF same-sample, logit clip .05 (new)	-0.011	0.179	0.179	0.980	0.746
DiD-BCF reference fold, RF (new)	-0.017	0.197	0.196	0.940	0.711
DiD-BCF reference fold, logit (new)	-0.052	0.558	0.558	0.880	0.954
TWFE	-0.022	0.198	0.198	0.980	0.894
Synthetic DiD	-0.024	0.165	0.166	0.990	0.792

Table 10: GATT($g=4, t=4$) in the information-ablation family (100 replications per cell). “reduced” uses only the two groups and periods of the cell; “pooled” fits one effect surface with `effect_by_cohort=False`.

Design	d	N	Variant	Bias	RMSE	Cov.95	Len.95
baseline	1	200	full panel	0.005	0.127	0.970	0.559
baseline	1	200	reduced cell	0.018	0.207	0.960	0.936
baseline	1	200	pooled	0.007	0.129	0.980	0.571
baseline	1	800	full panel	0.007	0.065	0.950	0.288
baseline	1	800	reduced cell	0.029	0.111	0.980	0.493
baseline	1	800	pooled	0.009	0.067	0.970	0.297
baseline	2	200	full panel	-0.034	0.137	0.980	0.574
baseline	2	200	reduced cell	-0.020	0.187	0.980	1.032
baseline	2	200	pooled	-0.035	0.140	0.970	0.590
baseline	2	800	full panel	0.013	0.065	0.970	0.295
baseline	2	800	reduced cell	0.013	0.097	1.000	0.521
baseline	2	800	pooled	0.014	0.066	0.990	0.305
serial	1	200	full panel	0.008	0.120	0.950	0.464
serial	1	200	reduced cell	-0.002	0.123	0.990	0.689
serial	1	200	pooled	0.009	0.124	0.930	0.476
serial	1	800	full panel	-0.005	0.058	0.950	0.234
serial	1	800	reduced cell	-0.001	0.060	1.000	0.460
serial	1	800	pooled	-0.004	0.059	0.970	0.242
serial	2	200	full panel	-0.015	0.134	0.940	0.484
serial	2	200	reduced cell	-0.023	0.126	0.990	0.734
serial	2	200	pooled	-0.017	0.135	0.930	0.497
serial	2	800	full panel	-0.002	0.075	0.890	0.249
serial	2	800	reduced cell	-0.006	0.074	1.000	0.477
serial	2	800	pooled	-0.002	0.076	0.910	0.259
staggered	1	200	full panel	-0.035	0.159	0.970	0.693
staggered	1	200	reduced cell	0.012	0.240	1.000	1.265
staggered	1	200	pooled	1.417	1.424	0.000	0.514
staggered	1	800	full panel	-0.007	0.097	0.940	0.389
staggered	1	800	reduced cell	0.022	0.118	1.000	0.668
staggered	1	800	pooled	1.467	1.469	0.000	0.260
staggered	3	200	full panel	-0.052	0.195	0.970	0.721
staggered	3	200	reduced cell	0.025	0.254	1.000	1.458
staggered	3	200	pooled	1.362	1.370	0.000	0.537
staggered	3	800	full panel	-0.012	0.093	0.980	0.414
staggered	3	800	reduced cell	0.019	0.134	0.990	0.732
staggered	3	800	pooled	1.425	1.427	0.000	0.275

Table 11: ATT in the information-ablation family (100 replications per cell). “reduced” uses only the two groups and periods of the cell; “pooled” fits one effect surface with `effect_by_cohort=False`.

Design	d	N	Variant	Bias	RMSE	Cov.95	Len.95
baseline	1	200	full panel	0.004	0.109	0.960	0.426

Design	d	N	Variant	Bias	RMSE	Cov.95	Len.95
baseline	1	200	pooled	0.006	0.106	0.970	0.427
baseline	1	800	full panel	0.007	0.054	0.940	0.215
baseline	1	800	pooled	0.007	0.054	0.940	0.215
baseline	2	200	full panel	-0.029	0.129	0.940	0.438
baseline	2	200	pooled	-0.030	0.129	0.940	0.440
baseline	2	800	full panel	0.014	0.058	0.920	0.226
baseline	2	800	pooled	0.015	0.057	0.930	0.227
serial	1	200	full panel	0.016	0.129	0.850	0.360
serial	1	200	pooled	0.017	0.132	0.850	0.361
serial	1	800	full panel	-0.005	0.062	0.880	0.183
serial	1	800	pooled	-0.004	0.062	0.830	0.182
serial	2	200	full panel	-0.009	0.146	0.770	0.382
serial	2	200	pooled	-0.008	0.149	0.770	0.383
serial	2	800	full panel	0.008	0.072	0.820	0.196
serial	2	800	pooled	0.009	0.071	0.850	0.196
staggered	1	200	full panel	-0.025	0.126	0.930	0.441
staggered	1	200	pooled	-0.224	0.266	0.550	0.476
staggered	1	800	full panel	0.011	0.055	0.950	0.219
staggered	1	800	pooled	-0.203	0.213	0.130	0.240
staggered	3	200	full panel	-0.049	0.141	0.860	0.468
staggered	3	200	pooled	-0.248	0.290	0.580	0.503
staggered	3	800	full panel	-0.007	0.057	0.960	0.235
staggered	3	800	pooled	-0.217	0.225	0.050	0.255

Table 12: Sharp null ($\tau \equiv 0$): rejection rate of the two-sided 5% posterior test. Published JBES run: 100 replications; new run: available replications.

Estimator	GATT(4, 4)				ATT			
	$d1, N200$	$d2, N200$	$d1, N800$	$d2, N800$	$d1, N200$	$d2, N200$	$d1, N800$	$d2, N800$
JBES plain (structured)	0.010	0.010	—	—	0.040	0.050	—	—
JBES corrected (same-sample)	0.090	0.080	—	—	0.060	0.040	—	—
New raw structured	0.040	0.060	0.020	0.010	0.030	0.070	0.080	0.030
New current hybrid, logit	0.060	0.050	0.060	0.060	0.060	0.030	0.060	0.000
New current hybrid, RF	0.070	0.040	0.040	0.030	0.070	0.030	0.060	0.020
New reference fold, logit	0.060	0.060	0.090	0.070	0.100	0.100	0.070	0.050
New reference fold, RF	0.050	0.060	0.090	0.090	0.090	0.080	0.090	0.040

Table 13: Head-to-head summary of the old and new corrections on GATT(4, 4): RMSE with 95% coverage in parentheses. The theory-aligned reference fold uses the random-forest pilot; “clip .05” is the same-sample logit with the clipping constant set to 0.05.

Cell	Old corrected	Same-sample logit	Same-sample clip .05	Reference fold RF
baseline, $d = 1, N = 200$	0.231 (0.95)	0.214 (0.96)	0.214 (0.96)	0.243 (0.91)
baseline, $d = 2, N = 200$	1.040 (0.96)	5.947 (0.98)	0.258 (0.98)	0.234 (0.97)
baseline, $d = 2, N = 800$	0.318 (0.99)	0.102 (0.96)	0.102 (0.96)	0.114 (0.95)
serial, $d = 1, N = 200$	0.136 (0.99)	0.145 (0.98)	0.145 (0.98)	0.160 (0.96)
serial, $d = 2, N = 200$	0.269 (0.99)	0.264 (0.99)	0.158 (0.99)	0.164 (0.98)

Table 14: Head-to-head summary against the published benchmarks on GATT(4, 4): RMSE with 95% coverage in parentheses. DiD-BCF entries are the new runs; benchmark entries are the published JBES runs.

Cell	Raw	Reference fold RF	Callaway–Sant’Anna	DoubleML	grf-DiD
baseline, $d = 1$, $N = 200$	0.131 (0.98)	0.243 (0.91)	0.221 (0.97)	0.304 (0.92)	0.222 (0.81)
baseline, $d = 2$, $N = 200$	0.135 (0.96)	0.234 (0.97)	0.225 (0.96)	0.280 (0.94)	0.223 (0.81)
baseline, $d = 1$, $N = 800$	0.069 (0.96)	0.130 (0.92)	0.107 (0.98)	0.114 (0.97)	0.111 (0.81)
serial, $d = 1$, $N = 200$	0.126 (0.92)	0.160 (0.96)	0.129 (0.99)	0.165 (0.99)	0.133 (0.91)
serial, $d = 2$, $N = 200$	0.141 (0.88)	0.164 (0.98)	0.135 (0.99)	0.170 (0.97)	0.133 (0.90)

References

- [1] H. G. Souto and F. Louzada. Forests for Differences: Robust Causal Inference Beyond Parametric DiD. arXiv:2505.09706, 2025. <https://doi.org/10.48550/arXiv.2505.09706>.
- [2] P. H. C. Sant’Anna and J. Zhao. Doubly robust difference-in-differences estimators. *Journal of Econometrics*, 219(1):101–122, 2020. <https://doi.org/10.1016/j.jeconom.2020.05.003>.
- [3] B. Callaway and P. H. C. Sant’Anna. Difference-in-Differences with multiple time periods. *Journal of Econometrics*, 225(2):200–230, 2021. <https://doi.org/10.1016/j.jeconom.2020.12.001>.